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Correction for the static shift in magnetotellurics using transient electromagnetic soundings

Ben K. Sternberg*, James C. Washburne*, and Louise Pellerin‡

ABSTRACT

Shallow inhomogeneities can lead to severe problems in the interpretation of magnetotelluric (MT) data by shifting the MT apparent resistivity sounding curve by a scale factor, which is independent of frequency on the standard log-apparent-resistivity versus log-frequency display. The amount of parallel shift, commonly referred to as the MT static shift, can not be determined directly from conventionally recorded MT data at a single site. One method for measuring the static shift is a controlled-source measurement of the magnetic field. Unlike the electric field, the magnetic field is relatively unaffected by surface inhomogeneities. The controlled-source sounding (which may be a relatively shallow sounding made with lightweight equipment) can be combined with a deep MT sounding to obtain a complete, undistorted model of the earth. Inversions of the static shift-corrected MT data provide a much closer match to well-log resistivities than do inversions of the uncorrected data.

The particular controlled-source magnetic-field sounding which we used was a central-induction Transient ElectroMagnetic (TEM) sounding. Correction for the static shift in the MT data was made by jointly inverting the MT data and the TEM data. A parameter which allowed vertical shifts in the MT apparent resistivity curves was included in the computer inversion to account for static shifts. A simple graphical comparison between the MT apparent resistivities and the TEM apparent resistivities produced essentially the same estimate of the static shift (within 0.1 decade) as the joint computer inversion.

Central-induction TEM measurements were made adjacent to over 100 MT sites in central Oregon. The complete data base of over 100 sites showed an average static shift between 0 and 0.2 decade. However, in the rougher topography and more complex structure of the Cascade Mountain Range, the majority of the sites had static shifts of the order of 0.3 to 0.4 decade. The static shifts in this area are probably due to a combination of topography and surficial inhomogeneities. The TEM apparent resistivity (which is used to estimate the unshifted MT apparent resistivity) does not necessarily agree with either the transverse electric (TE) or the transverse magnetic (TM) MT polarization. TEM apparent resistivity may occur between the two, or may agree with one of the two polarizations, or may lie outside the MT polarizations.

INTRODUCTION—MT STATIC PROBLEM

Field experience has shown that magnetotelluric (MT) apparent resistivity curves from nearby sites (as close as 100 m) generally have the same shape but are often shifted parallel to one another (that is, shifted along the vertical axis). These parallel shifts in the apparent resistivity curve can lead to large errors in the inverted models of the data. Inverted resistivities are shifted by a factor equal to the amount of parallel shift between the apparent resistivity curves, and inverted depths are shifted by an amount related to the square root of the resistivity shift. In order to interpret MT soundings correctly, it is necessary to remove these parallel shifts.

To illustrate the nature of these parallel shifts, a three-dimensional (3-D) body in a layered earth was modeled using an algorithm based on the method of integral equations (Wannamaker et al., 1984a). Figure 1 contains the plan, transverse, and cross-section views, which indicate the size and shape of the inhomogeneity. Both the surface inhomogeneity and the background layered-earth resistivity structure are representative of the survey area that is discussed as a case history in a later section. Figure 2 shows a plot for the Transverse

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*Department of Mining and Geological Engineering, Bldg. 12, College of Engineering and Mines, The University of Arizona, Tucson, AZ 85721.
‡Department of Geology and Geophysics, University of Utah, Salt Lake City, UT 84112.

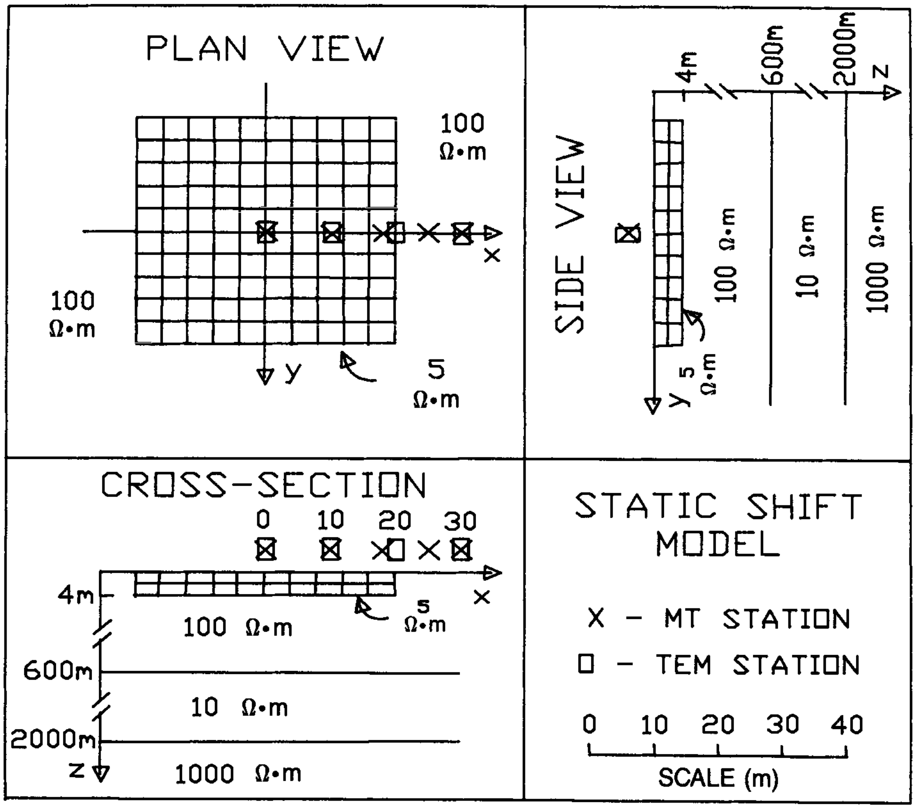

FIG. 1. Model used for calculation of static shift effect. A plan view (scale in lower right), as well as cross-section and transverse views, are shown. The 4 m thick surface inhomogeneity is imbedded in a three-layer earth. The layered earth is representative of the general resistivity section along the Transcascade survey line (Figure 7). The grid shows the discretization of the body into rectangular cells for the integral equation calculation.

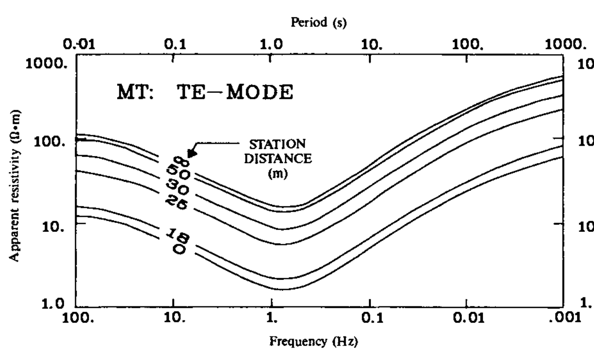

FIG. 2. Magnetotelluric apparent resistivity curves corresponding to the model in Figure 1 for Transverse Electric (TE) polarization. MT response was calculated by an integral equation method at the stations shown. Station distance noted on this plot is the distance from the center of the anomaly on Figure 1 to the MT station.

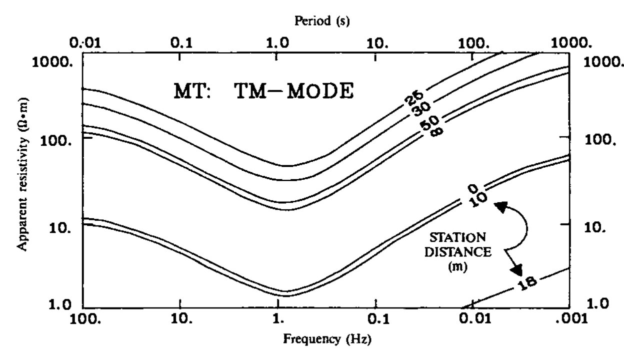

FIG. 3. Same as Figure 2 but for Transverse Magnetic (TM) polarization. Both the TE and TM modes result in widely shifted curves, depending on their positions relative to the inhomogeneity.

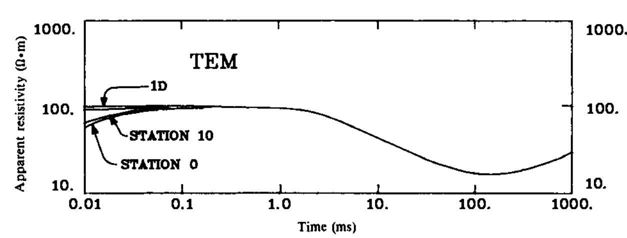

FIG. 4. Transient Electromagnetic (TEM) apparent resistivity curves corresponding to the model in Figure 1 with a 250 m loop. Response was calculated by an integral equation method. Station distance noted on this plot is the distance from the center of the anomaly in Figure 1 to the TEM station. Only at times less than 0.1 ms is there any significant variation from the layered-earth response.

Electric (TE) polarization of the theoretical MT apparent resistivity versus frequency for point receivers at sites along a profile line which crosses this surface inhomogeneity. Figure 3 shows a similar plot of the MT apparent resistivity versus frequency for the Transverse Magnetic (TM) polarization. At infinite distance from the inhomogeneity, the MT sounding is identical to an unperturbed, one-dimensional (1-D) sounding. At sites near the inhomogeneity, within about 50 m in this example, there is a large parallel displacement along the apparent resistivity axis for both the TE and TM polarizations. This shift is referred to as a static shift in analogy to the seismic static shift. The parallel shift is related to a distortion of the electric field caused by a build-up of boundary polarization charge at resistivity boundaries (Wannamaker et al., 1984b; Newman et al., 1986), an effect that has been variously referred to as current channeling or current gathering (Spies and Parker, 1984).

Figure 4 shows a plot of theoretical apparent resistivity curves for central-induction Transient ElectroMagnetic (TEM) soundings across the same surface inhomogeneity. The curve which is asymptotic to 100 $\Omega \cdot$m at early times is the undistorted, 1-D sounding curve. The other curves are for soundings at various distances from the center of the inhomogeneity. These TEM curves respond to the surface inhomogeneity only at very early times (less than 0.1 ms) and are undistorted and coincident with the 1-D sounding at times later than 0.1 ms. This behavior is to be expected, since the TEM sounding is a measurement of the magnetic field only and is not affected by the buildup of charge at the inhomogeneity boundaries.

The inhomogeneity in Figure 1 is representative of a large class of shallow inhomogeneities and surface topographic features which lead to charge buildup at boundaries and distort MT soundings. In contrast, the effect on TEM soundings (at relatively late times) is minimal.

The MT apparent resistivity calculations in Figures 2 and 3 assumed a point electric field dipole. The magnitudes of the calculated apparent resistivity anomalies would be reduced if the electric field dipole length were made large compared to the size of the surface inhomogeneity (Jones, 1988). Nevertheless, this reduction does not alter our conclusions, since our calculations are intended only to be representative of the effect of surface inhomogeneities. If we used a finite length dipole, we could still find a surface inhomogeneity of a certain scale which would produce anomalies similar to those in Figures 2 and 3.

The use of TEM soundings forms the basis for correcting the static shift in MT soundings. A relatively small-scale TEM sounding is made to a time which is large enough to provide resistivity information at depths below the surface inhomogeneity but not to such large depths that unwieldy loop sizes, very large transmitters, or excessive stacking times would be required. The MT soundings may be shifted vertically so that the high-frequency part of the MT curve agrees with the TEM curve. The low-frequency MT curve then provides an undistorted picture of the deep resistivity section. The amount of shift can also be determined by jointly inverting the TEM and MT apparent resistivity curves; the shift necessary to make the TEM and the MT models agree in the shallow layers is defined to be the static shift.

In this paper, we address only the problem of parallel static shifts caused by very shallow inhomogeneities. Larger scale inhomogeneities, on the order of a skin depth or larger, cause a frequency-dependent shift in the apparent resistivity sounding curve rather than a parallel shift. These 2-D and 3-D effects can be interpreted directly from the MT data using numerical modeling and are not discussed in this paper.

PREVIOUS WORK

Most previous work on static shift corrections for MT data falls in four groups. The following list is intended to be a representative but not an exhaustive review of previous work:

(1) Spatial filtering using closely spaced MT sites

Sternberg et al. (1982) determined a site-average apparent resistivity for each MT site and an area-average apparent resistivity surrounding each site. The apparent resistivity curve for each site was then shifted by the difference between the area average and the site average. An inversion based on the area-averaged apparent resistivity agreed with the well control much better than the unaveraged apparent resistivities. Warner et al. (1983) described a similar method which also averaged data from closely spaced sites. These averaging methods represent low-pass spatial filters.

Bostick (1986) and Torres Verdin (1985) described an enhanced spatial filtering method, which is called ElectroMagnetic Array Profiling (EMAP), in which the filter length varies with the depth of penetration. Data are collected with end-to-end electric field dipoles along a survey line which is oriented perpendicular to the presumed geoelectric strike. Full five-component MT stations are recorded at about every tenth electric dipole. The impedances for groups of sites are smoothed at each frequency with low-pass spatial filters that remove wavelengths that are less than a skin depth. Word et al. (1986) and Shoemaker et al. (1986) present several examples of EMAP processing.

A method related to spatial averaging and to EMAP is the use of very long electric field measuring dipoles in order to average the *E* field (Swift, 1967). All of these averaging techniques are most effective when the electric-field measuring dipole (or the equivalent dipole in EMAP) is large compared to the lateral extent of the shallow surface inhomogeneities. These techniques may not be effective when, for example, shallow surface inhomogeneities are aligned substantially parallel to the receiving array.

In order to use spatial filtering techniques effectively, there must be continuous (or at least closely spaced) sampling of the electric field along the profile line. Although spatial filtering of a profile line can reduce anomalies due to large static shifts, it may be prohibitively costly to record data at the necessary number of sites, particularly in large regional surveys. Even in detailed surveys, permit problems, terrain, or sources of electrical interference may prevent continuous sampling of the electric field.

(2) Theoretical calculations of the static shift from buried surface inhomogeneities

Berdichevsky and Dmitriev (1976a, b) discuss the effects of highly simplified surface inhomogeneities on MT sounding curves. Larsen (1977) calculated the local surface conductivity effects on mantle response curves. Recently there has been considerable interest in numerically calculating the MT response to 3-D near-surface bodies (Wannamaker et al., 1984b). With 3-D forward modeling, the theoretical MT response from arbitrarily shaped bodies can be calculated and, in principle, removed from the observed response.

Although calculations of the static shift due to simple bodies can provide considerable physical insight into the static shift problem, rarely is there sufficient information about the near-surface, often hidden, inhomogeneities within the survey area to define models adequately. Furthermore, the numerical calculations for arbitrarily shaped bodies are far too expensive in 1987 for routine application at each MT site.

(3) Theoretical calculations of the static shift from surface topographic effects

Since topography can be determined easily from standard topographic maps, the static shift due to topography can be accurately determined by theoretical calculations. Examples of these calculations are found in Jiracek and Holcombe (1981), Holcombe (1982), Reddig and Jiracek (1984), Reddig (1984), Jiracek et al., (1986), Mozley (1982), Chouteau and Bouchard (1986), and Wannamaker et al. (1986). Although static corrections based on topography have proven valuable, particularly in mountainous terrain, they do not solve static shift problems due to subsurface bodies. Substantial static shifts can and do occur in flat areas (Sternberg et al., 1982) due to shallow buried inhomogeneities.

(4) Interpretation based on known geology

For many years interpreters have dealt with static shifts as they have dealt with other complexities of real-world data: by applying their knowledge of the geologic framework of the region, by incorporating available well-log, gravity, magnetic, and seismic data, and by insight gained through experience in a particular area. Some areas with difficult static shift problems have been interpreted successfully using this approach.

Not surprisingly, two interpreters may estimate completely different static shifts at the same site using this approach. Clearly, objective evidence should be used for as much of the interpretation as possible. In a frontier area where MT is often applied, there may be little or no information available for such a subjective interpretation of static shifts. Of course, no matter what objective method of removing static shifts is used, the final interpretation must always take into account all available geologic and geophysical information to ensure that the solution is geologically reasonable and does not contradict known structural information.

Use of an independent meaurement

All of the above methods have made significant contributions toward solving the problem of static shifts in MT data. Nevertheless, there is the need for a method which directly estimates the amount of static shift using an independent measurement. Andrieux and Wightman (1984) presented results of using an EM37 time-domain controlled-source measurement to correct for the static shift. We wish to expand upon their results, particularly by comparing well logs to static shift-corrected MT inversions, in order to evaluate this technique.

Our goals in this paper are (1) to document the usefulness and limitations of static shift corrections using TEM data by comparing the shifted and unshifted MT data with well logs, and (2) to use a large data base consisting of combined MT and TEM measurements covering a wide variety of physiographic and geologic provinces to estimate the relative amounts and types of static shift that can occur in an MT survey. We shall not claim that the TEM method is the only viable static correction method. Indeed, in many cases various approaches are complementary.

PREFERRED APPROACH TO THE STATIC SHIFT PROBLEM—CENTRAL-LOOP TRANSIENT EM SOUNDING

Since the static shift is caused by boundary charges which mainly affect the electric field, measuring the magnetic field circumvents this problem. With a controlled-source method, where the transmitter signal strength is known, the apparent resistivity of the earth can be determined from just a single measurement of the magnetic field. Specifically, we used the Geonics EM37 time-domain recording system to make central-loop induction soundings. A large single-turn loop of wire was used as the transmitter. We experimented with three different transmitter loop sizes; the length of one side of the square loop was equal to 125 m, 250 m, or 500 m. Typically, the transmitter loop current was 15 to 20 A. A horizontal air-core loop with an effective area of 100 m^2 was used to record the time-derivative of the vertical component of the magnetic field at the center of the transmitter loop.

The transient decay waveform was recorded after turnoff of a transmitted pulse. Transmitter pulses with repetition frequencies of 30 Hz and 3 Hz were used. Data processing of these transient waveforms began by deconvolving the effect of the finite turn-off time (Levy, G.M., 1984, Geonics Ltd. Tech. note TN-16). The transient voltage data were then converted to apparent resistivities, using the nonlinear equation for EM fields due to a loop on a homogeneous half-space. This method determines, by an iterative procedure, the root (or resistivity) for the nonlinear half-space equation (Sternberg, 1977; Raab and Frischknecht, 1983; Spies and Eggers, 1986). No assumptions or approximations were made about late-time or early-time asymptotes as are required for explicit formulas of apparent resistivity for controlled-source EM soundings.

A sounding over a wide time range is required in order to provide an effective overlap of the controlled-source sounding region with the MT sounding; typically, a range of 100 μs to 100 ms was used. A joint inversion was then performed on both soundings with a parameter that allows the MT curve to shift along the vertical axis. Programs NLSTCI and IMSLPW (Anderson, 1979, 1982) were used for this purpose.

COMPARISON OF TEM AND MT APPARENT RESISTIVITY CURVES

An interpretation procedure that we have found to be quite useful, one which allows a direct comparison of the transient soundings with the MT soundings, involves the use of a simple time shift of the TEM sounding. After we divide the transient time scale (in ms) by 200 and compare the reduced scale with the MT time scale (in s), the TEM and MT apparent resistivity curves virtually overlap. In order to compare the MT apparent resistivity with the time-shifted transient apparent resistivity, the transient apparent resistivity must be obtained by an iterative solution of the nonlinear half-space equation. A late-time or early-time approximation to the apparent resistivity is not adequate.

An empirical justification for the validity of this simple time shift is provided by the results from 1-D model calculations (Figure 5). This figure shows calculated transient apparent resistivity and MT apparent resistivity curves with the time shift applied to the EM apparent resistivity curve. These four examples are representative of a much larger set of comparisons that we have made. As one can see from the three-layer models, the MT and TEM sounding curves are very similar after the time shift has been applied. If there were a static shift in the MT data, it could be determined directly by a comparison of the time-shifted TEM and MT data sets. For the two-layer models of Figure 5, the MT and TEM curves are quite similar for the smaller resistivity contrast case, but there is some disagreement along the steeply ascending portion of the curve for the larger resistivity contrast case. Only the near-horizontal parts of the curves should be used for estimation of static shift.

A somewhat different time shift may be more appropriate for a particular range of models. For example, an interpreter might use a slightly different time shift for each survey area that has a relatively consistent resistivity structure. Nevertheless, our experience indicates that, on the average, the factor of 200 shift seems to be reasonable.

A further justification for using a simple time shift to compare MT and TEM soundings is provided by comparing the depth of penetration of MT frequency-domain signals with the depth of penetration of transient signals. A commonly used definition of depth of penetration for EM fields is the skin depth, the distance in which the wave amplitude is reduced by $1/e$, or 37 percent.

In the frequency domain, the plane-wave solution to the diffusion equation for magnetic fields is given in many elementary texts as

$$H_y = H_0 \exp(-az) \exp[i(\omega t - az)],$$

where $a = \sqrt{\omega\mu\sigma/2}$, and H_0 is the incident magnetic field.

The second exponential represents simple harmonic motion with a phase shift and the first exponential represents the attenuation with distance. The skin depth, the distance z where $H_y/H_0 = 1/e$, is

$$z = \sqrt{2/(\sigma\mu\omega)}. \qquad (1)$$

In the time domain, for a step function excitation of magnitude H_0 established at time 0, the transient magnetic fields are (Nabighian, 1988)

$$h_y = H_0 \operatorname{erfc}[\sqrt{(\sigma\mu)/(2t)}\, z/\sqrt{2}].$$

The skin depth, the distance z where $h_y/H_0 = 1/e$, is

$$z = 1.28\sqrt{t/\sigma\mu}. \qquad (2)$$

If we equate the frequency-domain skin depth (1) to the time-domain skin depth (2), we have the relationship between MT frequency and transient time

$$194/f = t, \qquad (3)$$

where f is in Hz and t is in ms. The factor of 194 in equation (3) is remarkably close to the factor of 200 which was based on a comparison of a large number of layered-earth model calculations. Other definitions of depth of penetration could be used which would lead to somewhat different relationships than the factor of 200. Since all of these relationships are highly simplified approximations to the actual EM field behavior in an inhomogeneous earth, we will not explore the

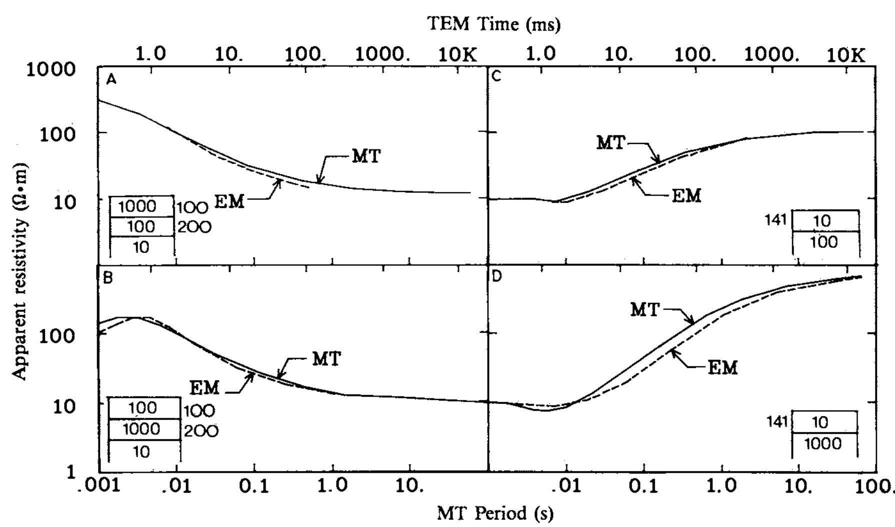

FIG. 5. Comparison of theoretical apparent resistivity curves for MT sounding (solid curve) and time-shifted EM sounding (dashed curve). The EM curve has been shifted by dividing the time axis (in ms) by 200. The radius of the circular transmitter loop that was used in this calculation was 141 m. The resistivity model used is shown in each panel.

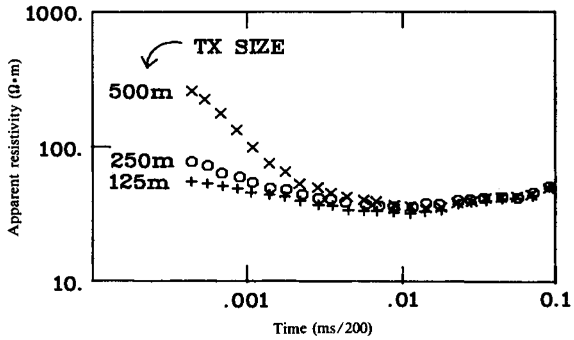

FIG. 6. Field measurements with varying loop sizes. The apparent resistivity curves are for square transmitter loops with sides of 125 m, 250 m, and 500 m. Except at early times, the curves are approximately coincident.

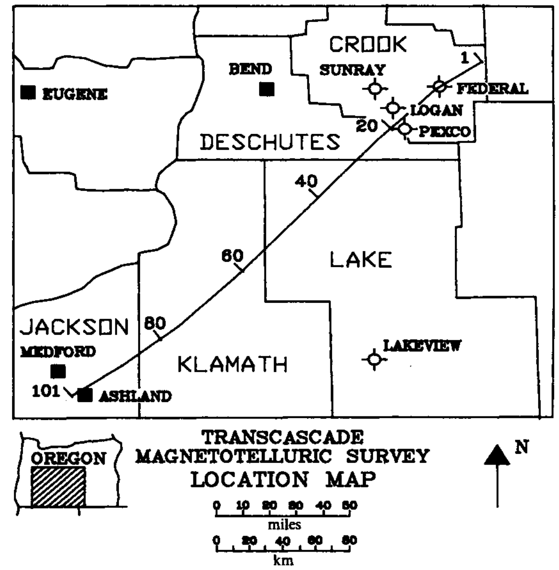

FIG. 7. Location of the Transcascade survey line in Oregon. In addition to the 101 sites spaced approximately equidistantly along the plotted line, four sites were placed near each of the wells in Crook County and one near a well in Lake County.

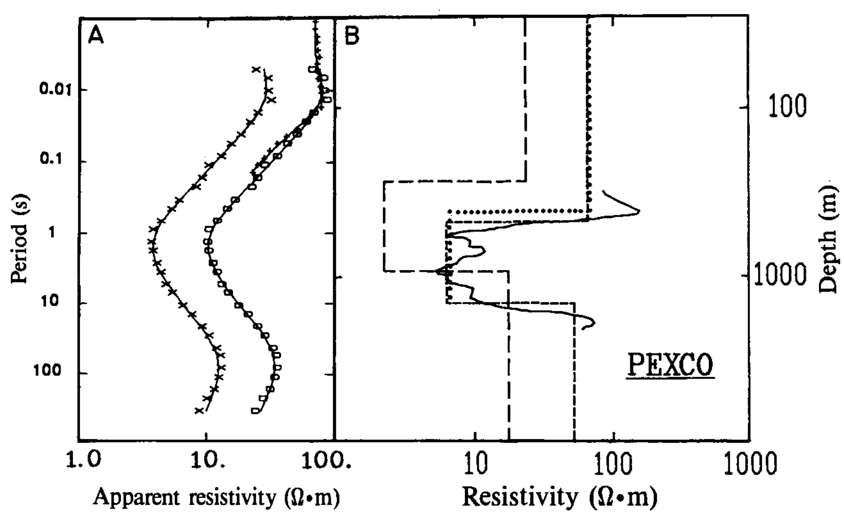

FIG. 8. (a) MT apparent resistivity curve for a site located near the Pexco well. The cross symbols are the recorded E polarization MT apparent resistivity sounding curve. The solid line through these symbols is the theoretically calculated MT apparent resistivity curve for the best-fit inverted model for these data. The open circle symbols are for the static shift-corrected apparent resistivity curve. The solid line through these symbols is the theoretically calculated MT apparent resistivity curve for the best-fit inverted model for these data. The plus symbols are the EM apparent resistivity sounding curve. (b) Well log and inversions of the apparent resistivity data. Solid line is the well log. Dotted line is the inverted model corresponding to the TEM data. Long dashed line is the inverted model corresponding to the unshifted MT data. Short dashed line is the inverted model corresponding to the shifted MT data. The amount of static shift that was applied to the MT data was obtained by jointly inverting the TEM and the MT data and determining the optimum shift that led to the same (or nearly the same) resistivity model in the shallow part of the section. The static shift was also determined by simply noting the vertical shift between the MT and the TEM curves on this plot. The two methods led to essentially the same estimate of the static shift (+0.44 decade).

relationships further. However, as long as we avoid comparing the steep parts of the TEM and MT curves, the exact factor that is used (e.g., 150 to 250) will not significantly alter the interpretation.

In support of our use of the simple time shift, we offer our own experience. We first determined static shifts by directly comparing the MT apparent resistivity curves with the time-shifted TEM apparent-resistivity curves. We then determined static shifts via a joint inversion using the complete equations for MT and TEM (for a layered earth). The two approaches led to the same static shifts to better than one-tenth decade accuracy at each site.

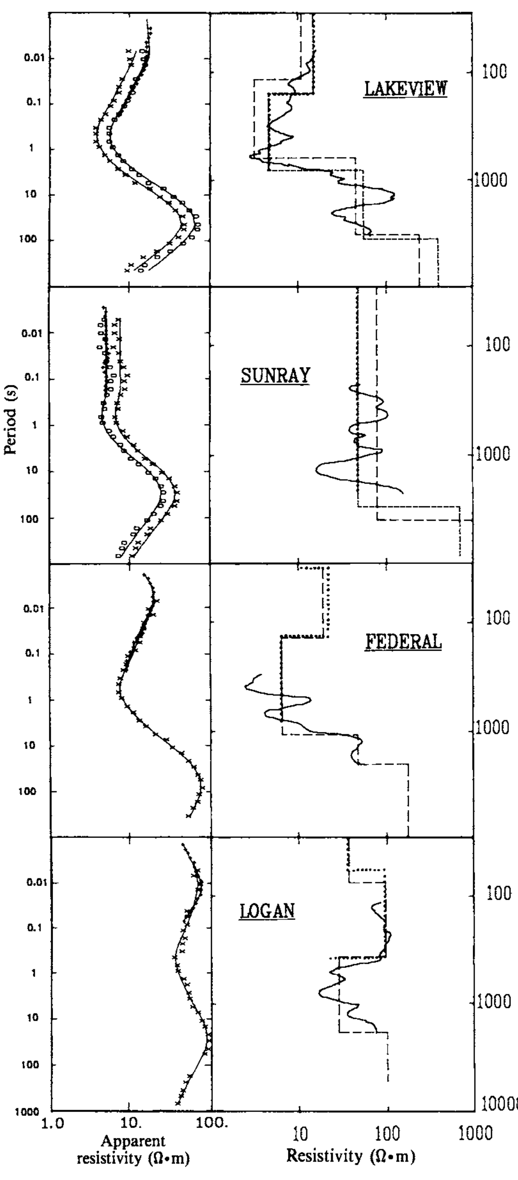

FIG. 9. Same as Figure 8 except it is for the Lakeview well site (+0.17 decade shift), Sunray well site (−0.17 decade shift), Federal well site (negligible static shift), and Logan well site (negligible static shift).

LOOP SIZE

A potentially important parameter in the central-induction TEM measurements is the size of the transmitting loop. Figure 6 is representative of experiments we have made with varying loop sizes. The three curves are different at short times but converge at later times, showing that the choice of loop size is not critical as long as relatively late-time apparent resistivities are used. This result is not surprising, since at late times, all loops act as dipoles and the loop size becomes immaterial (Spies and Frischknecht, 1988).

Experience from other surveys has shown that for small loops (100 m or smaller) the signal-to-noise ratio (S/N) is too low to let us reliably calculate late-time apparent resistivities. For very large loops (500 m or larger) the loop becomes unmanageable, and the loop may cross regional geologic contacts, complicating the interpretation. A compromise loop size of 250 m was chosen for the measurements reported below.

CASE HISTORY—TRANSCASCADE SURVEY

A location map for the Transcascade MT survey in central Oregon is shown in Figure 7. Approximately 100 sites were occupied along a 320 km (200 mile) long line. This large data set crossed several geologic provinces and therefore provides a good statistical basis from which to determine the average amount of static shift and the types of static shift that are likely to be encountered.

As a test of the TEM static shift correction method, we have used all available sites from this survey which had well logs available. These sites are Federal, Sunray, Logan, Pexco, and Lakeview (see Figure 7 for locations). At each of these sites, the MT station was within a few hundred meters of the well location, i.e., just far enough away to avoid interference from the steel casing in the well. The MT, TE polarization sounding curves for these five sites are shown in Figures 8 and 9. Both the observed data (symbols) and the calculated data based on the inverted model (solid line) are shown. In addition, the corresponding TEM data (+ symbols) are shown on this same plot. The time values for the TEM data have been divided by 200 before plotting as described in an earlier section of this paper.

In order to illustrate the effectiveness of this static shift correction, we have compared inversions of the static shifted and non-static shifted MT curves with the well-log models. Figure 8b shows the resistivity log from the Pexco well plotted on a logarithmic scale. Basically, the well log indicates a high-resistivity surface layer, followed by a conductive middle layer, and a resistive third layer. Also shown on Figure 8b are inversions of static-shift corrected and uncorrected MT curves. The corrected inversion consists of a simultaneous inversion of the TEM sounding and the MT sounding. The corrected inversion shows a resistivity model which is much closer to the well-log resistivities and to the depths from the well log than the model determined from the uncorrected MT curve. For example, the depth to the rapid increase in resistivity in the deep part of the well log ("basement") occurs at about 1700 m. The deep resistivity increase based on an inversion of the static shift-corrected MT curve occurs at 1600 m, i.e., within about 6 percent of the well-log depth. The deep resistivity increase based on an inversion of the non-static shift corrected MT curve occurs at 970 m, giving an error of about 40 percent. In addition, the middle conductive layer resistivity based on the inversion of the static shift corrected MT curve agrees very well with the average resistivity in the conductive middle section of the well log. The conductive layer resistivity from the inversion of the non-static shift corrected MT curve is much lower than the average resistivity in the well log as we would expect from the downward shift of the uncorrected MT curve. In summary, the correction of the MT sounding for static shift has provided a much more accurate picture of the resistivity structure of the earth.

In the Pexco example, the TE mode and the TM mode apparent resistivity curves were almost parallel at all frequencies. Furthermore, the TEM apparent resistivity curve agreed closely with the MT TM apparent resistivity curve. The success of static-shift correction at this site can be viewed as either (1) showing that a large correction to the TE mode is needed, or (2) a clear-cut signal to the interpreters that they should be using the curve that was picked as TM mode. In either case, the use of the TEM data leads to a far more accurate resistivity model at this site.

Figure 9 shows the simultaneous inversions for the other four sites. The static shift corrected inversions for both Sunray and Lakeview provide a closer match to the well logs than do the uncorrected inversions. At Federal and at Logan, there was good agreement between the original, uncorrected MT inversion and the well log. Nevertheless, these sites still provide a test of the EM static shift correction method. At both sites, the TEM data indicate no shift. This result provides evidence that the TEM data not only lead to valid static shift corrections but leave the data alone when they should.

It is also interesting to note that 1-D inversions provide a reasonably good picture of the gross resistivity structure at all five sites. After correction for the static shift caused by shallow 2-D and 3-D inhomogeneities, a 1-D inversion is often adequate to represent the resistivity structure. In other areas, a combined interpretation using multidimensional modeling of the large-scale inhomogeneities may be required along with static shift correction methods that handle the effects of the small-scale surface inhomogeneities.

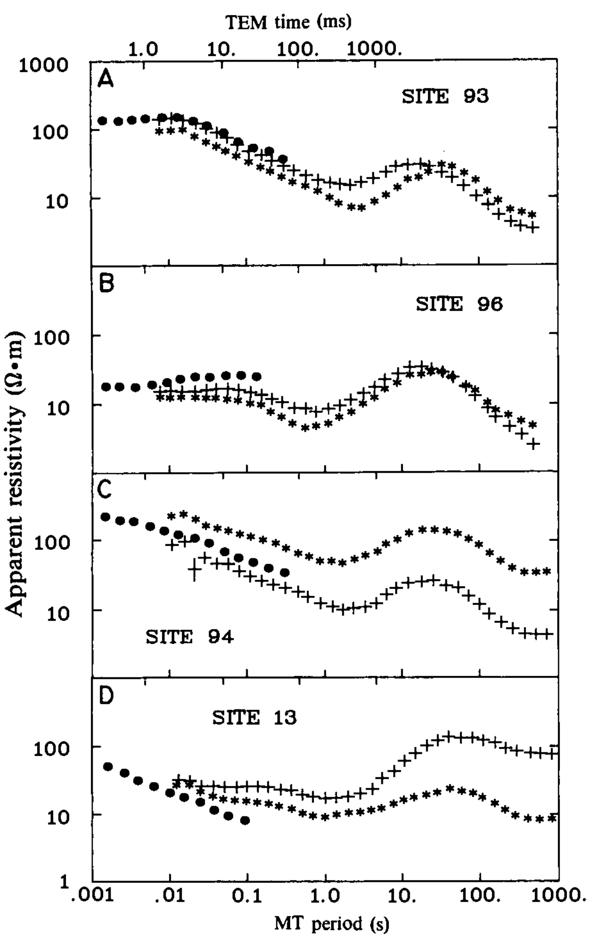

FIG. 10. (a) Comparison of the EM transient sounding curve (solid black dots) with the MT sounding curve at site 93 (star symbols are TE, plus symbols are TM). This example shows a close match between the EM transient sounding apparent resistivity curve and one of the two polarizations of the MT apparent resistivity curve. (b) Comparison of EM and MT sounding curves at site 96. In this case, the transient sounding curve is above both polarizations of the MT apparent resistivity curves. (c) Comparison of EM and MT sounding curves at site 94. The transient sounding curve is between the two MT polarizations. (d) Comparison of EM and MT sounding curves at site 13. The EM transient sounding curve shape is different from the MT curve shape where the two have overlapping data.

STATISTICS FOR THE NATURE AND AMOUNT OF THE STATIC SHIFT

It should be emphasized that a parallel shift between two sounding curves (TE and TM) at an MT site clearly indicates static shift effects. However, the absence of a shift between these curves may simply mean that both curves were static shifted the same amount, so a lack of shift does not guarantee there are no static problems. Furthermore, there is no reason to expect that either of the two MT polarizations will provide the correct resistivity. In general, the true resistivity may lie above, below, or between those from the two polarizations.

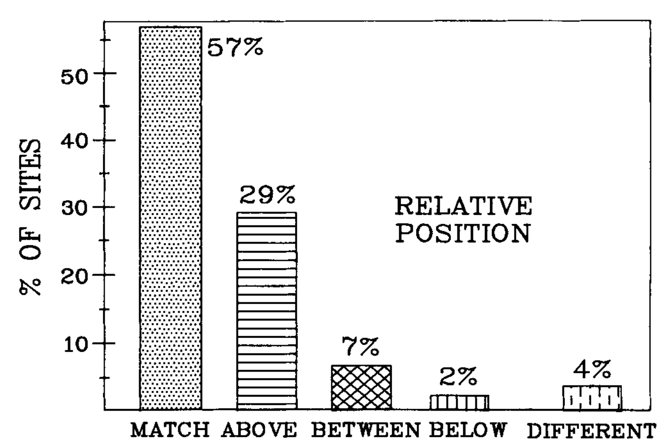

FIG. 11. Percentages of the various relative positions that were shown in Figures 10a, 10b, 10c, and 10d for all 101 survey sites.

Figures 2 and 3 illustrate the nature of shifts for both modes relative to the boundary of a surficial conductive inhomogeneity, i.e., when modes are shifted in the same direction and when they are shifted apart. These curves also show that the TM mode is more sensitive to surficial changes. Figure 10 illustrates these cases by comparing the transient apparent resistivity curve with the two MT curves (TE and TM polarization) for four sites from the Transcascade survey.

Figure 11 summarizes the relative percentages of matching for each of the four cases shown in Figures 10a, 10b, 10c, and 10d. Fifty-seven percent of the transient curves match (within ± 0.1 decade) one of the two MT polarizations. An MT interpreter will usually select one of the MT polarizations for analysis based on vertical magnetic field information or regional geologic information. If the selected polarization does not happen to be the one that agrees with the transient curve, some of the sites in this category could also have a significant static shift. Because of the uncertainties in selecting the appropriate curve for interpretation in this area, we have not broken down further the statistics for this category. Thirty-eight percent of the transient curves are either above, below, or between the two polarizations of the MT and therefore require a static shift correction. In 4 percent of the cases, the transient curve appears to have a substantially different shape than either MT curve, and therefore in 4 percent of the cases the TEM sounding does not provide a simple static shift correction. In other words, a joint inversion of the two data sets does not provide EM and MT models that are similar enough to allow us to estimate a static shift.

Of concern is not only the nature of the static shift but also its magnitude. Figure 12 summarizes the amount of static shift that occurred along the entire line. In 33 percent of the cases, there was essentially no static shift. In 18 percent of the cases, there was about 0.1 decade static shift. Depending on the noise level of the data and the geologic objectives, a static shift of less than 0.1 decade may be negligible. In 48 percent of the cases, there was a 0.2 to 0.8 decade static shift. Such large static shifts significantly affect layered-earth inversion models. Since the data set was quite large (approximately 100 sites), we feel that these are representative statistics for a fairly wide variety of geologic conditions. Considerable care was exercised during this survey to avoid obvious features that could cause static shifts, e.g., stream channels, hills, and roads. It is unlikely that additional field practices could reduce the average amount of static shift encountered on this survey.

The site location line is superimposed on the major geologic and physiographic provinces in the area in Figure 13. A major change occurred in the character of the static shift between those sites in the Basin and Range, the High Lava Plains, and the Blue Mountains and those sites that were in the Cascade Mountain Range. Figure 14 shows the statistics for the amount of static shift for those sites that were within the region labeled Cascade Mountain Range in Figure 13. Within this region, the average static shift is significantly larger; 58 percent of the sites have between 0.3 and 0.6 decade static shift. Here, only 23 percent of the cases show essentially no static shift and another 4 percent of the cases have 0.1 to 0.2 decade static shift. It appears that in the mountainous topography, complex structure, and high resistivity of the Cascade Mountain Range, we can expect to encounter considerably larger static shifts than over the survey as a whole.

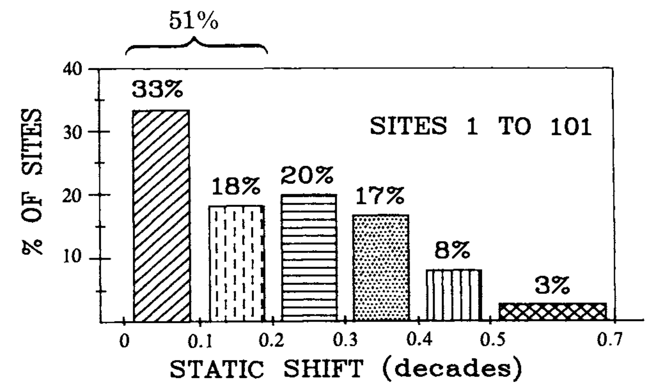

FIG. 12. Percentages of the relative amount of static shift for all sites along the profile line (sites 1 to 101). The amount of static shift is based on the fraction of a decade that the MT curve has shifted.

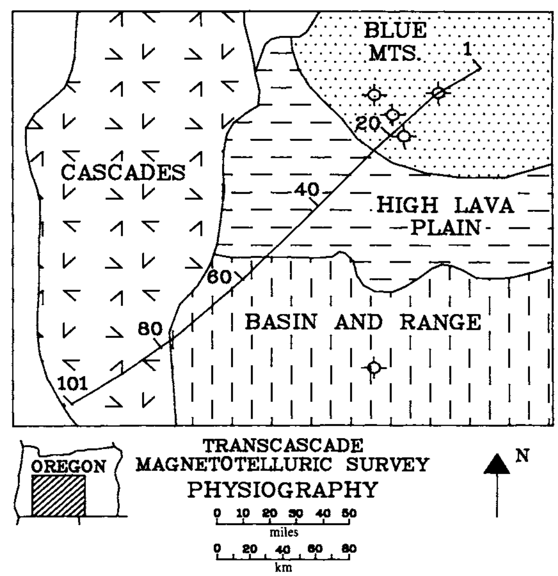

FIG. 13. Location of the MT sounding survey line superimposed on a generalized map of geology and physiography for the region.

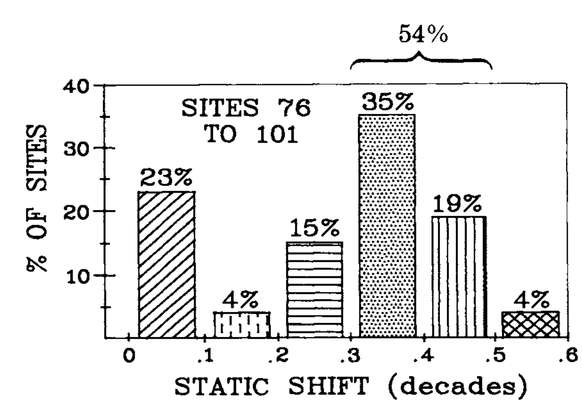

FIG. 14. The amount of static shift for sites 76 to 101, i.e., the sites within the Cascade Mountain Range, as shown on Figure 13.

CONCLUSIONS

We have found from a comparison of (a) static shift-corrected MT inversions, (b) non-static shift-corrected MT inversions, and (c) well logs at five sites that a correction for static shift by the use of controlled-source EM soundings in a joint inversion with the MT data can be very effective in providing a more accurate picture of the resistivity structure of the earth. Based on a data base of over 100 sites which crossed a number of geologic provinces, the average static shift in MT soundings was on the order of 0 to 0.2 decade. However, in areas of rough topography and complex structure such as in the Cascade Mountain Range, the majority of the sites had static shifts of the order of 0.3 to 0.5 decade. The corrected resistivity does not necessarily agree with either the minimum or the maximum apparent resistivity curve. Indeed, it may occur between the two, may agree with one of the two polarizations, or may lie outside the MT polarizations.

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